

Global Bread Intake (GBI) Study Global Dietary Database • Methodological Clarification Note	
Date:	29 April 2026
To:	GBI Study Data Contributors
From:	GBI Study Team (PI: Dr Renata Micha, Global Dietary Database)
Re:	Methodological Guidance on the Energy-Adjusted Bread Intake Method
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1. Purpose of this document

This methodological note clarifies the updated analytical approach for estimating energy-adjusted bread intake in the Global Bread Intake (GBI) Study. It is intended to support contributors who are preparing survey-level aggregate outputs and to ensure that energy-adjusted estimates are implemented consistently across surveys.

The updated guidance reflects a change in analytical priority. **The energy-adjusted method, particularly the residual method standardized to 2,000 kcal/day, is now the preferred approach.** Non-energy-adjusted estimates remain important as descriptive estimates of absolute intake as consumed. **Both unadjusted and energy-adjusted estimates are requested.** In the event of resource constraints that prevent contributors from providing both, **energy-adjusted estimates should be prioritized**, as these are the primary inputs for cross-survey comparability.

The document has the following objectives:

1. Outline the updated energy-adjusted method, including the residual approach standardized to 2,000 kcal/day and the rationale for key analytical choices.
2. Explain how the current specification differs from the previous instructions sent to data owners.
3. Provide step-by-step implementation guidance for the full energy-adjustment workflow, from person-level variable construction to stratum-level estimation.
4. Clarify the relationship between unadjusted and energy-adjusted estimates, including which is prioritized when resource constraints apply.
5. Specify the required deliverables and documentation expectations.

2. Background and rationale

In the original GBI instructions, contributors were asked to report non-energy-adjusted bread intake estimates for each bread definition, bread type, and population stratum. Energy-adjusted estimates were described as optional and were to be generated using the residual method standardized to a 2,000 kcal/day reference. The original wording allowed log transformation of bread intake depending on distributional assumptions and suggested adding a very small constant to retain individuals with zero intake when log transformation was required.

The updated specification changes this approach. **Energy-adjusted** estimates, standardized to **2,000 kcal/day**, are now the **preferred** analytical framework for comparison across strata and surveys. The updated method also clarifies how to construct person-level mean bread intake and person-level mean total energy intake, how to retain recall days with no bread consumption, how to treat non-consumers in the residual model, and how to report both unadjusted and energy-adjusted summary metrics.

3. What has changed compared with the previous instructions

The following tables summarize the main changes from the previous contributor instructions to the updated specification.

Component	Previous instruction	Updated specification
Preferred analytical approach	Non–energy-adjusted bread intake was presented as the primary requested output, while energy-adjusted estimates were optional.	The energy-adjusted residual method standardized to 2,000 kcal/day is now the preferred analytical approach for comparability across strata and surveys. Non–energy-adjusted estimates remain important and should also be reported alongside the energy-adjusted estimates where possible.
Age eligibility	Exclusion of participants aged <2 years was not explicitly highlighted.	Participants aged <2 years , if present, should be excluded from the analysis.
Averaging across recall days	Person-level intake was described, but the handling of multiple recall days was not fully detailed.	Person-level mean bread intake and person-level mean total energy intake must be calculated using the same recall-day structure per participant, based on the survey design. Person-level mean bread intake is computed by averaging across all valid recall days for the participant. Recall days with no bread consumption are retained as true zero-intake days (not treated as missing) and contribute to the across-day mean.
Same recall days for bread intake and energy intake	Not stated explicitly.	It is now clarified that the same available recall days per participant must be used when deriving person-level mean bread intake and person-level mean total energy intake.

Component	Previous instruction	Updated specification
Energy plausibility screening	Less explicitly integrated into the instructions.	Person-level mean daily energy intake should be screened using age- and sex-specific plausibility cut-offs. Contributors should determine, based on their survey context, whether to retain or exclude flagged participants from the analysis.
Handling missing energy	Not fully specified.	Participants with missing mean daily total energy intake are excluded only from energy-adjusted estimates (since the residual model requires both bread and energy values). They remain eligible for unadjusted estimates if they have valid bread intake data.
Handling of missing survey weights	Not specified.	If contributors choose to apply survey weights, weighted estimates (mean, SD, SE, proportion (%)) of non-consumers) should be calculated only among participants with non-missing weights .
Treatment of non-consumers in the residual model	Zero-intake participants could remain in the regression model through addition of a small constant (0.00001).	Participants with zero person-level mean bread intake (i.e., the across-day average is 0 g/day, occurring when bread intake is 0 g on every valid recall day) are excluded from the residual regression model . However, their adjusted bread intake is set to zero and they remain part of the survey-level descriptive estimation.
Transformation of variables	Not fully specified.	Both person-level mean bread intake and person-level mean total energy intake are log transformed (natural logarithm) before fitting the residual model.
Bread types adjusted in the model	Earlier wording emphasized total, refined, and whole-grain bread, but without complete clarification of modeling strategy.	Energy adjustment is performed separately for: • Bread alone: total bread • Bread alone: whole-grain bread • Bread alone: refined-grain bread • Bread from all sources: total bread • Bread from all sources: whole-grain bread • Bread from all sources: refined-grain bread Additivity (i.e. whole grain + refined grain = total) is not expected to hold in the energy-adjusted estimates for either bread definition (bread alone, bread from all sources). This is because the three bread types are estimated through three independent residual regression models, each with its own energy elasticity. Apparent discrepancies are a known and accepted feature of this approach and will be addressed downstream in the modelling stage. By contrast, additivity does hold for the unadjusted intakes by construction, since each bread item is classified as either whole grain or refined.

Component	Previous instruction	Updated specification
Final required outcomes		The following summary metrics for each stratum should be provided for each bread type: 1) Mean of energy-adjusted intake (g/day at 2,000 kcal/day) 2) SD of energy-adjusted intake (g/day at 2,000 kcal/day) 3) SE of energy-adjusted intake (g/day at 2,000 kcal/day) 4) Proportion (%) of non-consumers 5) Mean and SD of person-level mean total energy intake (kcal/day), reported per stratum If survey design parameters are available, they should be used in the estimation of the summary metrics described above. Crude metrics that ignore available design parameters should not be reported.
SD handling	Partition within- vs. between-person variance in each strata using analysis of variance (ANOVA).	For surveys with multiple recall/record days per participant, the SD reported should be the empirical standard deviation of the person-level mean intakes within the stratum. No statistical adjustment for within-person variability is required.
Documentation requirements	General documentation was expected.	A brief documentation note is required, covering the contributor's analytical choices and any deviations from this protocol (see Section 5 for details).

Note: The updated method no longer forces zero consumers into a log-scale model by adding a small constant. Instead, true zero-intake recall days are retained before person-level means are calculated, and participants with zero person-level mean bread intake are excluded only from model fitting. They return to the final survey-level dataset with adjusted intake set to zero.

4. Updated methodological specification

4.1 Scope of the updated method

The method should be applied separately for each available bread definition:

- **Bread alone**
- **Bread from all sources**

and separately for each bread type:

- **Total bread**
- **Whole-grain bread**

- **Refined-grain bread**

Energy-adjusted estimates should be generated at the **person-level first**, then summarized at the requested population or stratum level.

4.2 Step-by-step implementation

Step	Action	Methodological purpose
1	Confirm the analysis population	The energy-adjusted analysis should include eligible participants in the survey population. Participants aged <2 years , if present, should be excluded.
2	Identify all valid dietary recall or record days for each participant according to the survey design	Each participant may contribute one or more recall/record days. The number of days included should reflect the survey's dietary assessment design.
3	Calculate bread intake for each participant on each recall/record day	Bread intake should first be calculated at the recall-day level, separately for each bread definition and bread type.
4	Assign 0 g/day for recall days with no bread consumption	If a participant did not consume bread on a recall day, that day should be retained as a true zero-intake day, not treated as missing.
5	Calculate person-level mean bread intake	Average bread intake across all valid recall/record days for each participant, including zero-intake days.
6	Calculate total energy intake for each participant on each recall/record day	Energy intake should be calculated using the complete dietary record for each recall day.
7	Calculate person-level mean total energy intake	Average total energy intake across the same valid recall/record days used for bread intake.
8	Verify that the same recall-day structure is used for bread and energy	Person-level mean bread intake and mean energy intake should be based on the same recall days.
9	Identify participants with missing mean total energy intake	Exclude these participants only from energy-adjusted analyses.

Step	Action	Methodological purpose
10	Apply age- and sex-specific energy plausibility screening	Extreme energy intakes may indicate reporting or measurement problems. These values should be flagged, but not automatically excluded.
11	Use the recommended energy plausibility cut-offs	Suggested cut-offs are: for adults (≥ 20 years) females < 500 or $> 4,000$ kcal/day; males < 800 or $> 5,000$ kcal/day. For children and adolescents (2–19 years), age- and sex-specific cut-offs should be used (see Section 4.4).
12	Define consumers and non-consumers for each bread outcome	A participant is a non-consumer for a given outcome if their person-level mean intake for that outcome is 0 g/day — that is, the average across all valid recall/record days is 0, which occurs when bread intake is 0 g on every valid recall/record day. A consumer is any participant whose across-day average is > 0 (i.e., bread was consumed on at least one recall day).
13	Exclude non-consumers from the residual regression model	The residual model is fitted on the log scale and therefore includes only participants with non-zero person-level mean bread intake.
14	Log-transform bread intake and energy intake among consumers	Both person-level mean bread intake and person-level mean total energy intake should be log-transformed (natural logarithm) before modelling.
15	Fit the residual model among consumers	Regress log-transformed bread intake on log-transformed total energy intake. The regression should be fitted once per survey, pooled across strata (i.e. a single energy elasticity per bread outcome per survey), to ensure stable regression coefficients. Stratification (by age, sex, etc.) is applied only at the final summary stage.
16	Fit separate models for each bread outcome	Models should be run separately for: bread alone total, bread alone whole grain, bread alone refined grain, bread all sources total, bread all sources whole grain, and bread all sources refined grain.
17	Standardize intake to 2,000 kcal/day	Use the fitted model to estimate bread intake at a common reference energy level of 2,000 kcal/day. This is a statistical standardization for comparability across strata and surveys. Specifically, for each consumer i , compute the log-scale adjusted value as: $\log(\text{bread_i})_{\text{adj_i}} = \log(\text{bread_i}) - \beta \times [\log(\text{energy_i}) - \log(2,000)]$, where β is the estimated energy elasticity from the fitted regression.

Step	Action	Methodological purpose
18	Exponentiate adjusted values back to the original scale	Because the model is fitted on the log scale, adjusted values should be exponentiated to return to grams/day.
19	Assign adjusted intake of 0 g/day to non-consumers	Non-consumers are excluded from model fitting but should remain in the final population-level reporting dataset.
20	Apply survey design parameters, where available	If sampling weights are available, they should be applied at the stratum-level summary estimation , when computing the mean, SD, SE, and % non-consumers for each stratum, so that the resulting estimates are representative of the underlying population. The residual regression model itself (Step 15) is fitted unweighted, since the energy elasticity is treated as a structural relationship rather than a population parameter
21	Summarize estimates by requested population strata	After person-level adjusted intakes are created, estimate stratum-level summary statistics (mean, SD, SE, % non-consumers) based on the distribution of person-level intake values. The percentage of non-consumers in a stratum is defined as the percentage of participants in that stratum whose <i>person-level mean intake</i> (averaged across all valid recall/record days, including zero-intake days) is 0 g/day for the relevant bread outcome — that is, the average across all valid recall days is 0, which occurs when bread intake is 0 g on every valid recall day. This is therefore a person-level definition, not the percentage of recall days without bread consumption.
22	Report the required summary metrics	For each bread type and stratum, report: mean, SD, SE and % non-consumers for both the energy-adjusted intake (g/day at 2,000 kcal/day and the unadjusted intake (g/day). Also report mean and SD of person-level mean total energy intake (kcal/day) per stratum.
23	Document key methodological decisions	Transparent documentation is needed to interpret and compare estimates across surveys.

4.3 Treatment of non-consumers

Participants with **zero person-level mean bread intake** for a given bread outcome are excluded from the residual regression model for that outcome. This is because the model is fitted using log-transformed bread intake, and log transformation is not defined for zero values.

However, these participants should not be removed from the final population-level estimates. Their energy-adjusted intake should be set to 0 g/day for that outcome, and they should be included in the final calculation of survey-level or stratum-level summary estimates.

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4.4 Energy plausibility screening

Person-level mean total energy intake should be screened using age- and sex-specific plausibility cut-offs. Suggested cut-offs are shown in the table below.

Age group (years)	Sex	Lower (kcal/day)	Upper (kcal/day)
2–4	Both	< 300	> 2,000
5–9	Both	< 400	> 2,500
10–14	Both	< 500	> 3,500
15–19	Female	< 500	> 4,000
15–19	Male	< 800	> 5,000
≥ 20	Female	< 500	> 4,000
≥ 20	Male	< 800	> 5,000

These values are illustrative only. **Contributors should determine the cut-offs most appropriate for their population** and decide whether to retain flagged participants in the analysis or apply sensitivity analyses excluding them. Cut-offs may be informed by external standards or by the survey's own distribution of intake. The specific cut-offs applied and the decision regarding inclusion/exclusion of flagged participants should be documented. Participants outside these ranges should be flagged, but not automatically excluded from the primary analysis. The number and percentage of participants flagged should be documented. If contributors decide to conduct sensitivity analyses excluding implausible energy reporters, this should be clearly documented and reported separately.

5. Required outputs and reporting

For each bread definition, bread type, and population stratum, contributors should provide the following outputs:

Output	Required metrics	Notes
Energy-adjusted intake: 2,000 kcal/day	Mean, SD, SE, proportion (%) of non-consumers	Preferred residual-method standardization.
Unadjusted intake	Mean, SD, SE, proportion (%) of non-consumers	Absolute intake as consumed, g/day. Requested alongside the energy-adjusted estimates; if resource constraints prevent

Output	Required metrics	Notes
		providing both, energy-adjusted estimates take priority.
Total energy intake (per stratum)	Mean and SD of person-level mean total energy intake (kcal/day)	Reported per stratum, computed among participants with valid energy data. Supports interpretation of energy-adjusted bread estimates.
Documentation note	Brief note covering: (1) whether survey weights were applied; (2) energy adjustment method used; (3) any sensitivity analyses; (4) any survey-specific deviations.	Provided alongside the data deliverables, using the Survey_Info_Template (excel "GBI_Aggregate_Data_Form").

If survey design parameters are available, they should be used for estimation of these summary metrics. Crude metrics that ignore available design parameters should not be reported as the primary survey outputs.

6. Questions or clarifications

If you have questions about applying this guidance, please do not hesitate to contact the GBI Study team. We are happy to discuss individual cases.

Email: info@globalbreadintake.org
